

Karthik Bhattaram

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EDUCATION

University of California, Santa Barbara

Master of Science in Computer Science

Santa Barbara

Sept. 2026 to June 2027

- Concentration: Machine Learning, Network Security, Systems

University of California, Santa Barbara

Bachelor of Science in Computer Science

Santa Barbara

Sept. 2022 to June 2026

- Relevant Coursework: Machine Learning for Networking, Computer Security, Data Structures and Algorithms
- Honors: Regents Scholarship, College of Engineering Honors, USACO Silver Division

EXPERIENCE

Software Engineering Intern

Arista Networks

June 2025 – Sept. 2025

Santa Clara, California

- Created utilities to decode hardware registers specifying pause storm information in Ethernet ports
- Implemented a **PFC watchdog** that relayed the utility's output to a network switch's traffic manager

Machine Learning Researcher

University of California, Santa Barbara

Sept. 2024 – June 2025

Santa Barbara

- Developed pipelines to automate common network measurement efforts
- Curated network traffic datasets for a RAG-based **large language model**

Software Engineering Intern

Cadence Design Systems / OpenEye Scientific

July 2024 – Sept. 2024

Santa Fe, New Mexico

- Wrote **AWS**-based package to evaluate database compression algorithms
- Updated **CUDA C++** software using new language features to match competing solutions

Hardware Architecture Intern

Cadence Design Systems

June 2023 – Sept. 2023

San Jose, California

- Developed an automation tool to produce memory wrappers of various dimensions and architectures
- Collected **System Verilog** files defining memory modules to input to the tool

Software Engineering Intern

Axiado Corporation

June 2021 – Aug. 2022

San Jose, California

- Developed a package to create **network packet flows** based on IPv6, ARP, ICMP, etc. in simulation
- Deployed and expanded test suites of a hardware firewall's packet parser module

PROJECTS

College Course Database | *JDBC, SQL*

March 2025 – May 2025

- Designed a database schema using Entity-Relationship diagrams
- Wrote SQL commands to create the tables and enforce data integrity
- Developed a JDBC backend to handle database transactions
- Created two distinct command line interfaces with different permissions

TECHNICAL SKILLS

Certificates: Enterprise Security Professional, Data Science

Networking Technologies: Scapy, Wireshark, TCP/IP

Languages: Python, C/C++, Java, Shell, JavaScript, Verilog

Developer Tools: Git, Visual Studio Code, Vim, Emacs

Libraries: CUDA, Pandas, NumPy, PyTest, Black, Matplotlib